

## UST'-USA KOMI, Russian Federation

WMO: 234120

Lat: 65.9842N

Lon: 56.9058E

Elev: 260

StdP: 14.56

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -35.7      | -30.9 | -40.8                         | 0.5 | -35.7 | -35.7 | 0.7 | -30.7 | 19.5                  | 12.2 | 17.5 | 10.5 | 4.5                   | 20   | 0.756 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.4                   | 79.4            | 65.1 | 75.4 | 63.0 | 71.2 | 60.8 | 67.1                | 75.8 | 64.7 | 73.0 | 62.0 | 69.6 | 6.5                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 63.8                            | 89.5 | 71.9 | 61.1 | 81.1 | 69.0 | 58.4 | 73.7 | 66.6 | 31.8          | 75.9 | 29.9 | 73.1 | 28.0 | 69.5 | 75.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 18.6              | 16.5 | 14.7 | -38.5                      | 84.0 | 7.0                | 3.7 | -43.5                                              | 86.7 | -47.6      | 88.9 | -51.5      | 90.9 | -56.6      | 93.7 |
|                   |      |      | -38.6                      | 70.1 | 6.9                | 2.8 | -43.6                                              | 72.1 | -47.6      | 73.8 | -51.5      | 75.3 | -56.6      | 77.4 |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 28.2   | -0.6  | 1.6   | 14.0  | 25.6  | 37.8  | 51.1  | 58.7  | 52.5  | 43.5  | 31.1  | 14.5  | 6.9   |
|                                                                               | DBStd       | 23.42  | 16.70 | 16.80 | 13.10 | 11.13 | 10.91 | 9.85  | 9.07  | 7.19  | 6.86  | 8.59  | 14.55 | 17.17 |
|                                                                               | HDD50       | 8563   | 1569  | 1355  | 1116  | 731   | 410   | 107   | 22    | 53    | 214   | 585   | 1064  | 1337  |
|                                                                               | HDD65       | 13487  | 2034  | 1775  | 1581  | 1181  | 845   | 428   | 241   | 392   | 644   | 1050  | 1514  | 1802  |
|                                                                               | CDD50       | 613    | 0     | 0     | 0     | 0     | 30    | 139   | 292   | 132   | 20    | 0     | 0     | 0     |
|                                                                               | CDD65       | 64     | 0     | 0     | 0     | 0     | 1     | 11    | 46    | 6     | 0     | 0     | 0     | 0     |
|                                                                               | CDH74       | 474    | 0     | 0     | 0     | 0     | 7     | 88    | 344   | 35    | 0     | 0     | 0     | 0     |
|                                                                               | CDH80       | 77     | 0     | 0     | 0     | 0     | 0     | 11    | 62    | 4     | 0     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 7.7    | 7.3   | 7.4   | 8.0   | 7.9   | 8.3   | 8.1   | 7.4   | 7.1   | 7.3   | 8.0   | 7.5   | 7.5   |
| Precipitation                                                                 | PrecAvg     | 20.0   | 1.3   | 1.0   | 0.9   | 1.0   | 1.5   | 1.8   | 2.4   | 2.5   | 2.3   | 2.2   | 1.6   | 1.5   |
|                                                                               | PrecMax     | 27.7   | 2.4   | 2.0   | 1.8   | 2.1   | 2.9   | 4.2   | 5.0   | 6.5   | 4.6   | 4.2   | 3.0   | 2.4   |
|                                                                               | PrecMin     | 12.8   | 0.6   | 0.1   | 0.3   | 0.3   | 0.2   | 0.4   | 0.5   | 0.5   | 0.4   | 0.6   | 0.8   | 0.3   |
|                                                                               | PrecStd     | 3.5    | 0.4   | 0.4   | 0.5   | 0.5   | 0.7   | 0.9   | 1.0   | 1.3   | 0.9   | 0.8   | 0.6   | 0.5   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 32.4  | 33.8  | 42.4  | 55.0  | 74.0  | 81.5  | 85.3  | 78.7  | 66.1  | 52.2  | 37.9  | 34.1  |
|                                                                               |             | MCWB   | 31.4  | 32.7  | 36.0  | 44.1  | 56.8  | 64.9  | 68.2  | 67.5  | 57.4  | 48.4  | 36.9  | 33.1  |
|                                                                               | 2%          | DB     | 28.5  | 30.6  | 35.8  | 46.9  | 66.8  | 76.6  | 81.5  | 72.4  | 60.6  | 48.1  | 34.9  | 32.4  |
|                                                                               |             | MCWB   | 27.6  | 29.4  | 32.3  | 39.4  | 53.9  | 61.6  | 66.5  | 62.8  | 54.8  | 45.7  | 34.2  | 31.6  |
|                                                                               | 5%          | DB     | 25.1  | 27.6  | 33.4  | 42.9  | 60.9  | 71.5  | 78.0  | 68.2  | 56.6  | 44.6  | 33.2  | 30.6  |
|                                                                               |             | MCWB   | 24.3  | 26.4  | 31.2  | 37.1  | 50.4  | 59.6  | 64.7  | 60.3  | 51.9  | 42.8  | 32.4  | 29.7  |
|                                                                               | 10%         | DB     | 21.0  | 23.7  | 31.3  | 39.5  | 55.4  | 67.2  | 73.9  | 64.5  | 53.4  | 41.9  | 31.8  | 27.8  |
|                                                                               |             | MCWB   | 20.1  | 22.6  | 29.6  | 35.5  | 47.1  | 57.0  | 62.7  | 58.7  | 50.2  | 40.6  | 31.1  | 26.9  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 31.6  | 33.0  | 36.6  | 44.8  | 58.7  | 66.4  | 71.0  | 68.7  | 59.1  | 49.4  | 37.2  | 33.3  |
|                                                                               |             | MCDB   | 32.4  | 33.8  | 41.2  | 54.1  | 70.4  | 77.6  | 81.2  | 77.4  | 64.2  | 51.0  | 37.7  | 33.7  |
|                                                                               | 2%          | WB     | 27.8  | 29.6  | 33.1  | 40.5  | 54.7  | 64.0  | 68.6  | 64.6  | 55.4  | 45.8  | 34.2  | 31.9  |
|                                                                               |             | MCDB   | 28.5  | 30.4  | 35.0  | 45.8  | 65.6  | 73.0  | 77.8  | 70.0  | 59.9  | 47.9  | 34.6  | 32.3  |
|                                                                               | 5%          | WB     | 24.4  | 26.4  | 31.7  | 38.0  | 51.2  | 61.1  | 66.3  | 61.4  | 52.7  | 43.0  | 32.6  | 29.7  |
|                                                                               |             | MCDB   | 25.1  | 27.4  | 33.4  | 41.6  | 59.9  | 69.5  | 74.7  | 66.3  | 55.6  | 44.3  | 33.0  | 30.5  |
|                                                                               | 10%         | WB     | 20.1  | 22.7  | 29.4  | 35.9  | 47.4  | 58.1  | 64.0  | 58.8  | 50.6  | 40.7  | 31.1  | 26.9  |
|                                                                               |             | MCDB   | 20.9  | 23.7  | 31.2  | 39.1  | 55.2  | 66.3  | 72.7  | 63.9  | 53.1  | 41.9  | 31.7  | 27.8  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 11.7  | 11.8  | 14.8  | 14.5  | 14.3  | 14.6  | 15.4  | 12.4  | 9.9   | 7.0   | 9.4   | 10.8  |
|                                                                               |             | MCDBR  | 11.6  | 10.6  | 10.7  | 15.6  | 21.6  | 20.6  | 20.1  | 19.0  | 15.1  | 8.1   | 7.4   | 9.8   |
|                                                                               |             | MCWBR  | 11.5  | 10.1  | 8.7   | 10.5  | 12.1  | 10.8  | 10.0  | 11.1  | 9.6   | 6.9   | 7.5   | 10.0  |
|                                                                               | 5% WB       | MCDBR  | 11.5  | 10.2  | 10.2  | 13.2  | 21.1  | 18.6  | 18.6  | 16.5  | 13.3  | 7.9   | 7.2   | 10.0  |
| MCWBR                                                                         |             | 11.4   | 9.8   | 8.8   | 9.6   | 12.8  | 11.6  | 10.3  | 11.2  | 9.6   | 6.9   | 7.3   | 10.2  |       |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.221 | 0.272 | 0.309 | 0.339 | 0.348 | 0.356 | 0.393 | 0.370 | 0.330 | 0.290 | 0.215 | 0.223 |
|                                                                               | taud        |        | 1.882 | 2.022 | 2.063 | 2.101 | 2.316 | 2.455 | 2.353 | 2.391 | 2.469 | 2.262 | 1.932 | 1.902 |
|                                                                               | Ebn at Noon |        | 61    | 178   | 232   | 257   | 270   | 271   | 254   | 249   | 235   | 177   | 61    | 6     |
|                                                                               | Edh at Noon |        | 6     | 19    | 30    | 38    | 34    | 30    | 33    | 29    | 21    | 15    | 5     | 2     |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 30    | 212   | 683   | 1217  | 1589  | 1597  | 1564  | 1012  | 544   | 185   | 48    | 5     |
|                                                                               | RadStd      |        | 2     | 26    | 80    | 115   | 131   | 153   | 236   | 113   | 66    | 25    | 8     | 0     |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)  | (e)                    | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) | Station Only           | +2.30 | N/A     | N/A    | N/A     | N/A   | N/A   | -731        | -806  | N/A   | N/A   |
| (47) | Regional (4 neighbors) | +2.11 | N/A     | N/A    | N/A     | N/A   | N/A   | -702        | N/A   | N/A   | N/A   |

Nomenclature: See separate page